Estimating the Hubble Constant: From Historical to Modern Data

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Contents

1 Introduction **2**

2 Part 1: Reproducing Hubble's Original Analysis (1929) **3**
2.1 Data and Method . . . 3
2.2 Results . . . 4
2.3 Discussion . . . 4
2.4 Partial Conclusion . . . 5

3 Part 2: Modern Distance Estimates (NED) **5**
3.1 Data and Method . . . 5
3.2 Results . . . 7
3.3 Weighted Fit . . . 7
3.3.1 A check on errors in both variables (ODR) . . . 10
3.3.2 Statistical error versus choice of method . . . 10
3.3.3 Validation against the published MCP result . . . 10
3.4 Discussion . . . 11
3.5 Limitations . . . 12

4 Conclusion **14**

A NED Query Details **14**

References **15**

Abstract

This project estimates the Hubble constant H_0 using two independent datasets: Edwin Hubble's original 1929 sample of 24 extragalactic objects, and a modern sample of 39 galaxies with distances and redshifts drawn from the NASA/IPAC Extragalactic Database (NED) and, for the six galaxies with geometric maser distances, from the Megamaser Cosmology Project. By comparing historical and modern results, the project examines how limitations in the extragalactic distance scale of the 1920s led to a significant overestimate of the expansion rate of the Universe, and situates both results against current measurements on both sides of the Hubble tension. Statistical uncertainties are attached to every fit and cross-checked with jackknife and bootstrap resampling. A sensitivity analysis shows that the six maser-distance galaxies leave the central value of the modern estimate essentially unchanged but reduce its uncertainty by a factor of about 2.6. An inverse-variance weighted fit, combining representative distance-method uncertainties with a peculiar-velocity scatter floor, illustrates a common pitfall (naively weighting by distance precision alone reintroduces the very peculiar-motion bias the sample extension was designed to suppress), and an orthogonal distance regression additionally allows for measurement error on distance. A paired bootstrap shows that the spread among the ten resulting estimators (70.4–80.4 km/s/Mpc) is comparable to their own sampling noise: no estimator differs from the reference fit by more than 1.8 times the noise of that difference. Finally, the weighted method is validated against the published Megamaser Cosmology Project result, which it reproduces to within 0.4σ.

1 Introduction

Hubble's Law is one of the foundational results of modern cosmology: it provided the first empirical evidence that the Universe is expanding, and remains central to how astronomers measure cosmic distances, ages, and the overall scale of the Universe. The value of the Hubble constant H_0 sets the present-day expansion rate of the Universe and is directly linked to its estimated age: a higher H_0 implies a younger Universe, and vice versa. Hubble's Law states that the recession velocity v of a galaxy is proportional to its distance r from us:

$$v = H_0\, r \tag{1}$$

where H_0 is the Hubble constant, typically expressed in km/s/Mpc.

It is worth being explicit about what Equation 1 actually is: the leading-order (linear) term of a Taylor expansion of the cosmic scale factor $a(t)$ around the present time, valid only for redshifts $z \ll 1$. Writing the expansion one order further introduces the deceleration parameter q_0, so that the distance-redshift relation gains a quadratic correction, $cz \approx H_0 r \left[1 + \frac{1}{2}(1 - q_0)\, \frac{H_0 r}{c} + \dots\right]$, which becomes relevant only once recession velocities are a non-negligible fraction of the speed of light. Every object in both samples used here has $v \ll c$ (at most about 3.4% of c for the most distant maser galaxy, NGC 6264, at $v \approx 10{,}145$ km/s), so the linear, single-parameter form used throughout this project is an excellent approximation, and no attempt is made to constrain q_0 or any higher-order cosmological parameter from this data.

This project revisits the very first measurement of H_0, published by Edwin Hubble in 1929 [Hubble, 1929], and compares it with a modern estimate obtained using the same statistical method but with contemporary distance measurements. The goal is not simply to reproduce a historical number, but to understand *why* Hubble's original estimate differed so substantially from the value accepted today, and what this reveals about the evolution of observational astronomy over the past century. Hubble's own initial value, $H_0 \approx 500$ km/s/Mpc, was already revised upward within a few years, and in 1935 Arthur Eddington, combining relativity and quantum theory, confidently claimed to have found "a solution which I regard as definitive," with $H_0 = 865$ km/s/Mpc [Tully, 2023] — a reminder that overconfidence in a single method has a long history in this field, one this project tries to avoid by explicitly comparing two independent approaches, attaching uncertainties to every result, and being transparent about their respective limitations.

Despite nearly a hundred years of increasingly precise measurements, the exact value of H_0 remains a subject of active debate. Measurements based on the local Universe — using distance-ladder methods such as Cepheids and Type Ia supernovae — give $H_0 = 73.3 \pm 1.0$ km/s/Mpc [Riess et al., 2022], while measurements inferred from the cosmic microwave background under the standard ΛCDM model give $H_0 = 67.4 \pm 0.5$ km/s/Mpc [Planck Collaboration, 2020]. This discrepancy, known as the *Hubble tension*, is quoted by the SH0ES team at more than 5σ significance — meaning that the probability of it being merely a statistical fluctuation would be less than about 1 in 3.5 million. Whether it signals new physics beyond ΛCDM or unrecognized systematic errors in the local distance ladder is still debated. The most recent analysis of the Chicago-Carnegie Hubble Program (CCHP), which calibrates Type Ia supernovae with the Tip of the Red Giant Branch (TRGB) using both the Hubble and the James Webb Space Telescopes, finds an intermediate value, $H_0 = 70.39 \pm 1.22\ (\text{stat}) \pm 1.33\ (\text{sys}) \pm 0.70\ (\sigma_{\text{SN}})$ km/s/Mpc, which its authors describe as consistent with ΛCDM without the need for new physics [Freedman et al., 2025], a conclusion that remains disputed. While a full treatment of this tension is beyond the scope of this project, it highlights that the precise determination of H_0 remains an open problem at the forefront of astrophysics — one to which this project's own estimates, despite their modest scale, can be compared, provided their uncertainties are stated honestly.

2 Part 1: Reproducing Hubble's Original Analysis (1929)

2.1 Data and Method

I used the original dataset published by Hubble (1929), consisting of 24 extragalactic objects with measured distance r (in megaparsecs) and radial velocity v (in km/s). To estimate the Hubble constant, I performed a linear fit of the form $v = H_0 r$, using two approaches: an unconstrained linear regression (allowing a free intercept) and a regression forced through the origin, which is the physically motivated form of Hubble's Law, since a galaxy at zero distance should show zero recession velocity due to cosmic expansion. For every fit, I also report the standard error on H_0 from ordinary least squares and the coefficient of determination R^2, to quantify both the precision and the quality of the fit. Table 1 reports the full dataset used, taken directly from Hubble's original 1929 paper: distance r (in Mpc) and radial velocity v (in km/s) for each of the 24 objects in the sample.

Object	r (Mpc)	v (km/s)
S. Mag.	0.032	+170
L. Mag.	0.034	+290
N.G.C. 6822	0.214	−130
N.G.C. 598	0.263	−70
N.G.C. 221	0.275	−185
N.G.C. 224	0.275	−220
N.G.C. 5457	0.45	+200
N.G.C. 4736	0.5	+290
N.G.C. 5194	0.5	+270
N.G.C. 4449	0.63	+200
N.G.C. 4214	0.8	+300
N.G.C. 3031	0.9	−30
N.G.C. 3627	0.9	+650
N.G.C. 4826	0.9	+150
N.G.C. 5236	0.9	+500
N.G.C. 1068	1.0	+920
N.G.C. 5055	1.1	+450
N.G.C. 7331	1.1	+500
N.G.C. 4258	1.4	+500
N.G.C. 4151	1.7	+960
N.G.C. 4382	2.0	+500
N.G.C. 4472	2.0	+850
N.G.C. 4486	2.0	+800
N.G.C. 4649	2.0	+1090

Table 1: Distance and radial velocity data for the 24 objects in Hubble's (1929) sample.

2.2 Results

Sample	Free intercept (km/s/Mpc)	Forced through origin (km/s/Mpc)
Full sample (24 galaxies)	454 ± 75 $(R^2 = 0.62)$	424 ± 42 $(R^2 = 0.62)$
Excluding Local Group (18 galaxies)	409 ± 98 $(R^2 = 0.52)$	433 ± 41 $(R^2 = 0.52)$

Table 2: Estimated H_0 from Hubble's 1929 data under different fitting assumptions, with standard errors and R^2 from ordinary least squares.

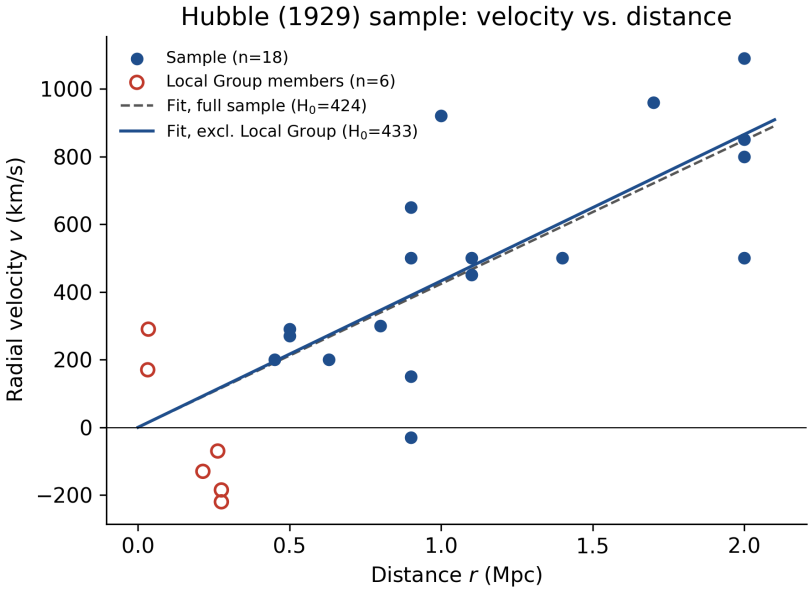


Figure 1: Velocity vs. distance for Hubble's 1929 sample, with linear fit forced through the origin.

For comparison, the currently accepted value of H_0 is approximately 67–73 km/s/Mpc [Riess et al., 2022, Planck Collaboration, 2020], meaning Hubble's original estimate overstated the expansion rate by a factor of roughly six, a discrepancy more than four standard errors wide even by the generous uncertainty of the 1929 fit itself.

2.3 Discussion

Several of the nearest objects in the sample — including the Magellanic Clouds, NGC 6822, M33, M32, and M31 (Andromeda) — belong to the Local Group, the small cluster of galaxies gravitationally bound to the Milky Way. At these short distances, gravitational interactions between galaxies (peculiar motion) dominate over the recession caused by cosmic expansion. This is directly visible in the data: M31 shows a negative velocity (v = -220 km/s), meaning it is approaching us rather than receding, alongside NGC 3031, which shows a similar effect. Excluding these six Local Group members shifts the two fits in opposite directions: the free-intercept slope decreases (454 → 409 km/s/Mpc), while the origin-forced slope increases slightly (424 → 433 km/s/Mpc). Both shifts are comparable to, or smaller than, one standard error on the respective fit, so this asymmetry is only mildly suggestive rather than a strong effect; it is nonetheless consistent with the origin-forced fit being less sensitive to the distortion introduced by nearby peculiar motions, and is therefore treated here as the more reliable estimator in this context. Notably, even after removing the Local Group, H_0 remains around 409–433 km/s/Mpc

— still roughly six times higher than the modern value, and far outside the uncertainty of either fit. This indicates that peculiar motion alone cannot account for Hubble's overestimate. The dominant source of error was a systematic underestimate of the distances themselves, which the following decades corrected in several steps rather than all at once.

It helps first to recall how Hubble's distances were obtained [Hubble, 1929]. Only the first seven objects — the six Local Group members and NGC 5457 — had distances based on Cepheids and other recognizable stars. The next thirteen assumed that the brightest stars in every galaxy share a common upper limit of luminosity, $M_{\rm pg} \approx -6.3$, calibrated on the nearby galaxies; the last four, members of the Virgo Cluster, were placed at a common distance of about 2 Mpc. Of the 18 objects that remain after excluding the Local Group, therefore, only NGC 5457 has a Cepheid-based distance.

The first correction came from Walter Baade's discovery of two populations of Cepheids with different luminosities [Baade, 1956]: the Cepheids in Hubble's calibrating galaxies are about 1.5 mag brighter, at a given period, than the calibration assumed, which increased extragalactic distances by about a factor of 2 [Tully, 2023] and would therefore roughly halve H_0. Because the brightest-star criterion was calibrated on the same nearby galaxies, this correction propagated to the whole sample — but it accounts for only about a factor of 2 out of the factor of six. The second step came from Humason et al. [1956], who showed that many of the objects taken as the "brightest stars" in more distant galaxies were in fact HII regions, clouds of ionized gas around groups of hot stars, intrinsically much brighter than any single star; with this and the Cepheid correction they obtained $H_0 \approx 180$ km/s/Mpc. Two years later, Sandage [1958] revised the brightest-star calibration further and placed H_0 between about 50 and 100 km/s/Mpc, with a preferred value of 75 [Tully, 2023]. Hubble's factor of six is therefore the product of a chain of errors in the distance scale — the Cepheid zero point first, then the misidentification of HII regions as stars and the recalibration of the brightest-star method — and not of the Cepheid calibration alone. Unlike the statistical uncertainties quoted above, none of these errors can be corrected after the fact within this analysis, since they affect the distances themselves.

2.4 Partial Conclusion

Hubble's 1929 estimate of H_0 was overstated primarily because his distances were systematically too small, rather than because of the specific choice of galaxies in the sample or of the fitting method. The underestimate came from a chain of errors in the distance scale of the time: the Cepheid zero point, corrected by Baade, accounts for about a factor of 2, and most of the remainder came from the later revision of the brightest-star distances, which were the majority of the sample, beginning with the recognition that many of Hubble's "brightest stars" were HII regions. This sets up a natural comparison with modern distance measurements in Part 2, using the NED catalogue [NED, n.d.].

3 Part 2: Modern Distance Estimates (NED)

3.1 Data and Method

To compare Hubble's 1929 result with a modern estimate obtained using the same statistical approach, a sample of 39 galaxies outside the Local Group was selected, using distance and redshift data retrieved from the NASA/IPAC Extragalactic Database (NED). For each galaxy, the distance r (in Mpc) was taken from the "Distances" entry, preferring Cepheid-based measurements where available. For galaxies without Cepheid data (mostly ellipticals), Surface Brightness Fluctuations (SBF) were used instead, and for the most distant part of the sample, geometric maser distances from the Megamaser Cosmology Project (MCP) [Pesce et al., 2020] were used, together with one Type Ia supernova (SNIa), one Type II supernova (SNII), and one Tip of the Red Giant Branch (TRGB) distance where no better method was available. These methods, together

with their respective systematic uncertainties, are reviewed in detail in Freedman and Madore [2010]. Local Group members were deliberately excluded, applying the same methodological lesson drawn from Part 1. The sample was deliberately extended to include five MCP maser galaxies (UGC 3789, NGC 6264, NGC 5765b, NGC 6323, and CGCG 074-064), which reach distances of 51.5–132.1 Mpc and provide some of the most precise, geometric distance measurements available, independent of the traditional distance ladder. Together with NGC 4258, whose geometric maser distance anchors the distance ladder, they form the six maser-distance galaxies in the sample. NED lists several distance estimates for NGC 4258, including values of about 6.4 and 7.6 Mpc; the maser distance of 7.58 Mpc [Reid et al., 2019] was adopted, as the most precise available and the one on which modern Cepheid calibrations are anchored. The same criterion — adopting the most recent and precise published determination — was applied to the five MCP galaxies, whose distances are taken from the combined MCP analysis [Pesce et al., 2020]. That analysis re-modelled four of them (UGC 3789, NGC 6264, NGC 6323, and NGC 5765b) with revised position-error floors, superseding the earlier individual determinations (49.6, 144, 107, and about 126 Mpc respectively), and it also lists a geometric maser distance for CGCG 074-064 (87.6 Mpc), used here instead of the Fundamental Plane distance (74.4 Mpc) adopted in earlier versions of this project. For transparency, this update alone raised the origin-forced estimates by 2.4 and 2.5 km/s/Mpc (version 1.2.0 of this analysis relative to version 1.1.0, which used the earlier maser distances and gave 71.7 and 73.2 km/s/Mpc for the two velocity frames).

Five further distances were corrected in version 1.2.1 after checking them against the literature. NGC 4594 (the Sombrero galaxy) had been listed as a Cepheid distance of 6.22 Mpc, but no Cepheid distance exists for this galaxy: the value matches the lowest globular-cluster luminosity function estimate in the published range (6.22–21.7 Mpc; McQuinn et al. 2016), and it is replaced by the TRGB distance of 9.55 ± 0.13 (stat) $\pm$ 0.31 (sys) Mpc [McQuinn et al., 2016]. For NGC 300, the adopted 1.49 Mpc was well below the Cepheid measurements, and it is replaced by the near-infrared Cepheid distance of 1.88 Mpc ($\mu_0 = 26.37$) of Gieren et al. [2005]. For NGC 2403, NGC 3351, and NGC 3627, the adopted distances (2.88, 8.99, and 8.73 Mpc) were 0.08–0.18 mag shorter than the final HST Key Project Cepheid moduli, 27.48, 29.85, and 29.86 mag (without metallicity correction; Freedman et al. 2001), and are replaced by the corresponding 3.13, 9.33, and 9.38 Mpc. Because these galaxies are all within 10 Mpc, where the fit gives them little weight, the five corrections change the unweighted full-sample estimates by at most 0.22 km/s/Mpc; the remaining Cepheid distances were not re-sourced (Section 3.5). Table 3 reports the full sample used.

Two velocity frames were tested: the heliocentric radial velocity $v_{\rm helio}$, and the velocity corrected for the Sun's motion relative to the cosmic microwave background, $v_{\rm CMB}$, both taken directly from NED. As in Part 1, two linear fits of the form $v = H_0 r$ were performed for each velocity frame: one with a free intercept, and one forced through the origin, again with standard errors and R^2 computed for each.

Object	r (Mpc)	Method	$v_{\rm helio}$ (km/s)	$v_{\rm CMB}$ (km/s)
NGC 4258	7.58	Maser	461	667
NGC 3368	9.72	Cepheids	888	1237
NGC 3627	9.38	Cepheids	722	1070
NGC 4321	14.2	Cepheids	1571	1896
NGC 4535	14.8	Cepheids	1964	2299
NGC 4536	12.2	Cepheids	1808	2151
NGC 4548	14.3	Cepheids	486	808
NGC 1365	17.2	Cepheids	1636	1539
NGC 1425	20.9	Cepheids	1512	1412
NGC 2403	3.13	Cepheids	133	182
NGC 300	1.88	Cepheids	147	−88
NGC 3351	9.33	Cepheids	779	1128
NGC 3982	20.5	Cepheids	1122	1291

Object	r (Mpc)	Method	$v_{\rm helio}$ (km/s)	$v_{\rm CMB}$ (km/s)
NGC 4496A	13.6	Cepheids	1730	2072
NGC 4639	19.7	Cepheids	1025	1345
IC 4182	4.07	Cepheids	321	550
NGC 5253	3.01	Cepheids	407	681
NGC 7331	14.4	Cepheids	816	490
NGC 4038	18.1	Cepheids	1624	1979
NGC 5236	4.5	Cepheids	513	793
NGC 4594	9.55	TRGB	1089	1433
NGC 1097	24.1	SNII (optical)	1271	1105
NGC 2841	13.1	Cepheids	635	806
NGC 3379	8.8	SBF	907	1255
NGC 3115	8.51	SBF	686	1041
NGC 1023	9.6	SBF	636	432
NGC 4494	15	SBF	1342	1634
NGC 5866	14.7	SBF	755	825
NGC 4278	14.9	SBF	649	934
NGC 7619	53	SBF	3762	3391
NGC 3384	9.46	SBF	704	1051
NGC 4589	20.4	SBF	1980	2030
NGC 2986	29.7	SNIa	2302	2637
NGC 1400	16.3	SBF	590	467
UGC 3789	51.5	Maser	3185	3245
NGC 6264	132.1	Maser	10145	10145
NGC 5765b	112.2	Maser	8257	8465
NGC 6323	109.4	Maser	7695	7662
CGCG 074-064	87.6	Maser	6915	7175

Table 3: Distance, measurement method, heliocentric velocity, and CMB-frame velocity for the modern galaxy sample (Local Group excluded, $n = 39$). Distances of the maser galaxies are from Reid et al. [2019] (NGC 4258) and Pesce et al. [2020] (the five MCP galaxies), those of NGC 300, NGC 2403, NGC 3351, NGC 3627, and NGC 4594 from Gieren et al. [2005], Freedman et al. [2001], and McQuinn et al. [2016] (see text); all other values are from NED.

3.2 Results

Fit type	H_0, $v_{\rm helio}$ (km/s/Mpc)	H_0, $v_{\rm CMB}$ (km/s/Mpc)
Free intercept	73.4 ± 2.1 $(R^2 = 0.972)$	72.0 ± 2.7 $(R^2 = 0.951)$
Forced through origin	74.1 ± 1.6 $(R^2 = 0.972)$	75.7 ± 2.2 $(R^2 = 0.945)$

Table 4: Estimated H_0 from the extended modern sample ($n = 39$, Local Group excluded), using heliocentric vs. CMB-frame velocities, with standard errors and R^2 from ordinary least squares.

3.3 Weighted Fit

The fits above treat every galaxy as equally informative, regardless of how precisely its distance is known. As a first attempt at correcting this, each galaxy was assigned a representative fractional distance uncertainty based on its distance method, using the typical ranges summarized in Freedman and Madore [2010]: 7% for Cepheids, 10% for SBF, 8% for SNIa, 15% for SNII (optical), and 5% for masers; the single TRGB distance is given 5%, slightly more than its own quoted uncertainty of about 3.5% [McQuinn et al., 2016]. These are method-level, representative

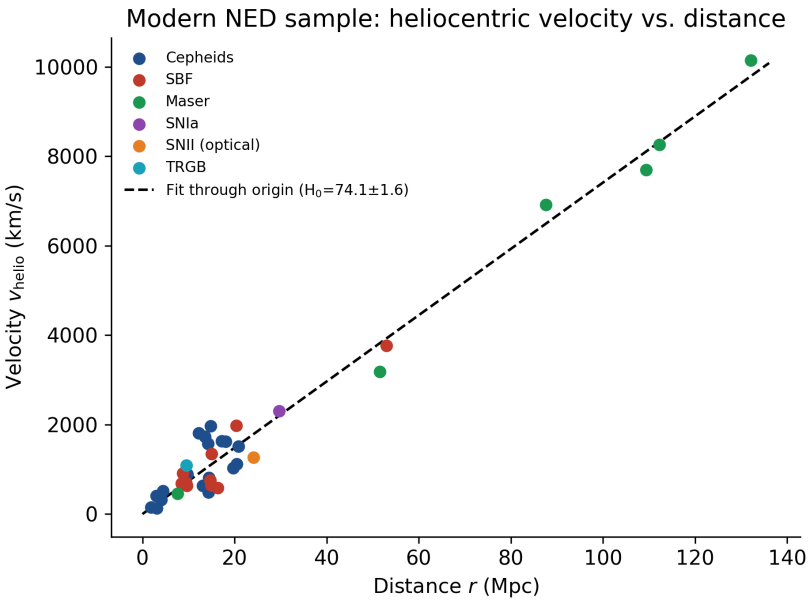


Figure 2: Heliocentric velocity vs. distance for the full modern sample ($n = 39$), including the Megamaser Cosmology Project galaxies out to 132 Mpc, with linear fit.

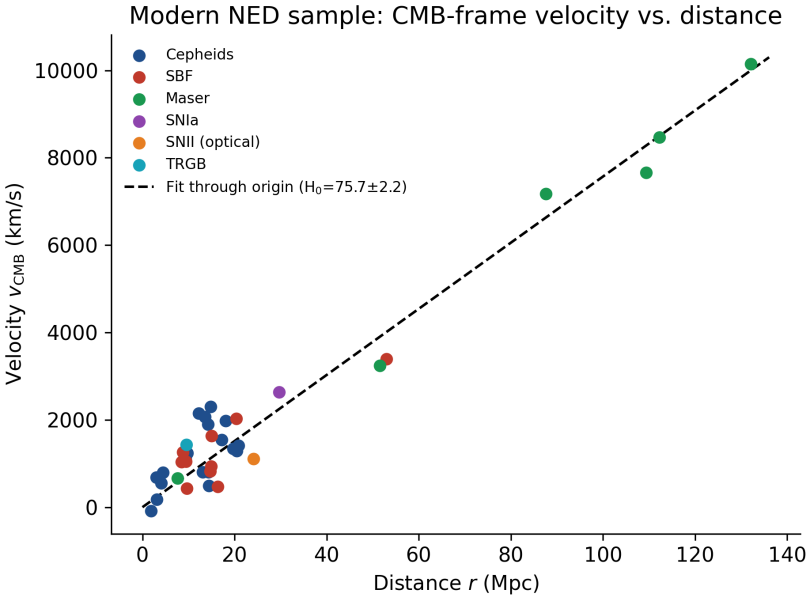


Figure 3: CMB-frame velocity vs. distance for the full modern sample ($n = 39$), including the Megamaser Cosmology Project galaxies out to 132 Mpc, with linear fit.

values, not individual NED-quoted errors for each object, and should be read as an order-of-magnitude correction rather than a precise error budget.

A first, naive attempt is instructive precisely because it fails: weighting each point by the distance-implied uncertainty on v alone ($\sigma_v = H_0 \cdot f_{\rm method} \cdot r$, propagated through an iteratively reweighted least-squares scheme) pulls the origin-forced fit to $H_0 = 80.6 \pm 4.4$ km/s/Mpc for $v_{\rm helio}$ and 94.8 ± 7.8 km/s/Mpc for $v_{\rm CMB}$ — both well outside the accepted range, with a poor fit quality ($\chi^2/{\rm dof} = 24$ and 55 respectively). The reason is instructive: because $\sigma_v \propto r$ in this naive scheme, nearby galaxies are assigned artificially small uncertainties and therefore dominate the fit, even though it is precisely at small r that peculiar motions (not distance-measurement error) are the dominant source of scatter. In effect, naive distance-only weighting reintroduces the same low-r, peculiar-velocity-dominated bias that the sample extension described in Section 3.1 was designed to suppress.

A more defensible weighting scheme therefore adds a distance-independent peculiar-velocity floor $\sigma_{\rm pec}$ in quadrature, $\sigma_v = \sqrt{(H_0 \cdot f_{\rm method} \cdot r)^2 + \sigma_{\rm pec}^2}$, reflecting the fact that peculiar velocities of a few hundred km/s do not shrink simply because a galaxy's distance happens to be well measured. Adopting a representative $\sigma_{\rm pec} = 200$ km/s (consistent with typical galaxy peculiar velocities of order 100–300 km/s relative to the Hubble flow) gives:

Fit type	H_0, $v_{\rm helio}$ (km/s/Mpc)	H_0, $v_{\rm CMB}$ (km/s/Mpc)
Free intercept	71.9 ± 3.5 ($\chi^2/{\rm dof} = 3.0$)	70.4 ± 4.5 ($\chi^2/{\rm dof} = 5.0$)
Forced through origin	74.3 ± 2.7 ($\chi^2/{\rm dof} = 2.9$)	78.9 ± 3.8 ($\chi^2/{\rm dof} = 5.6$)

Table 5: Weighted H_0 estimates using inverse-variance weights combining method-level distance errors with a $\sigma_{\rm pec} = 200$ km/s peculiar-velocity floor.

The heliocentric forced-origin estimate (74.3 ± 2.7) is essentially unchanged from the unweighted result (74.1 ± 1.6), while the CMB-frame estimate shifts upward, from 75.7 ± 2.2 to 78.9 ± 3.8, by less than one standard error of the weighted fit. As shown in Section 3.3.2, shifts of this size between estimators are comparable to the estimators' own sampling noise. The result is also reasonably stable to the exact choice of $\sigma_{\rm pec}$: repeating the fit with $\sigma_{\rm pec} = 150$ and 300 km/s instead of 200 gives $H_0 = 74.8$ and 74.1 km/s/Mpc for $v_{\rm helio}$, and 80.6 and 77.3 km/s/Mpc for $v_{\rm CMB}$, respectively — the same qualitative picture in all three cases. In every case, $\chi^2/{\rm dof}$ remains above 1 (between about 1.5 and 9, depending on frame and $\sigma_{\rm pec}$), which is expected: a single isotropic peculiar-velocity floor is itself a simplification, since real peculiar motions are direction-dependent (e.g. infall toward the Virgo Cluster or the Great Attractor) rather than a uniform scatter term, and a fully professional treatment would model these flows explicitly rather than absorb them into a single number.

As a complementary check, the unweighted fractional residuals, $(v - H_0 r)/(H_0 r)$, were also examined by method. Below, each group mean is quoted with its standard error ($s/\sqrt{n}$, where s is the scatter of the group), and the scatter is given separately. The 20 Cepheid-based points have a positive mean residual of $+0.16 \pm 0.10$ (scatter 0.45), 1.6 standard errors from zero: they lie on average above the fit, as expected for nearby galaxies, whose velocities are strongly affected by local peculiar motions — two of the three largest residuals (+0.8 to +1.0) belong to the Virgo Cluster galaxies NGC 4535 and NGC 4536 — and possibly also because some of their distances are still somewhat short (Section 3.5). In version 1.2.0 this mean was $+0.24 \pm 0.11$ ($n = 21$), 2.2 standard errors from zero, with the largest single residual (+1.36) coming from NGC 4594, which was then mislabelled as a Cepheid galaxy at too short a distance. The 10 SBF points are consistent with zero (-0.04 ± 0.10, scatter 0.30), and the 6 masers are both centered near zero and far tighter than either (-0.05 ± 0.04, scatter 0.10). The single SNIa, SNII, and TRGB points cannot support a statistical statement on their own. This is consistent with, though not strong independent proof of, the geometric maser distances being the most internally precise

method in the sample — exactly why Section 3.4 below finds them so influential.

3.3.1 A check on errors in both variables (ODR)

Every fit so far, weighted or not, is a form of ordinary least squares: it assumes the distance r is known exactly and puts all the measurement error on the velocity v. This is not quite right, since r is, if anything, the less certain of the two quantities. Orthogonal Distance Regression (ODR) relaxes this assumption: instead of minimizing vertical distances to the fit line, it minimizes a weighted combination of the horizontal and vertical distances, using an error budget on *both* axes. On the distance axis this is $\sigma_r = f_{\rm method} \cdot r$; on the velocity axis it includes only the terms that genuinely belong to the velocity, namely the $\sigma_{\rm pec} = 200$ km/s peculiar-velocity floor and a representative measurement error of 5 km/s, added in quadrature. ODR itself propagates the distance error onto the velocity axis, so the effective variance of each point is $(H_0 f_{\rm method} r)^2 + \sigma_{\rm pec}^2 + \sigma_{\rm meas}^2$, the same combination used in the weighted fit. Version 1.1.0 of this analysis mistakenly also included the distance term $H_0 f_{\rm method} r$ in σ_v, counting the distance error twice. This under-weighted the distant galaxies, where the distance error dominates, relative to the nearby ones, where peculiar velocities dominate — the same kind of distortion that affects the naive weighting above — and biased the ODR estimates upward: with the present data, the double-counted version would give 75.5 and 82.4 km/s/Mpc instead of the corrected values below.

The corrected ODR gives $H_0 = 75.0 \pm 2.7$ km/s/Mpc for $v_{\rm helio}$ (compared to 74.3 ± 2.7 for the weighted fit) and $H_0 = 80.4 \pm 4.0$ km/s/Mpc for $v_{\rm CMB}$ (compared to 78.9 ± 3.8). The direction of the change is the expected one: ignoring measurement error on the independent variable causes ordinary least squares to systematically *underestimate* the slope of a line, an effect statisticians call regression dilution or attenuation bias, so allowing for it pushes H_0 up. Here the effect is modest (0.7 and 1.5 km/s/Mpc), well within one standard error.

3.3.2 Statistical error versus choice of method

The five estimators applied to each velocity frame (ordinary least squares with a free intercept or through the origin, the two weighted fits, and ODR) span 71.9–75.0 km/s/Mpc for $v_{\rm helio}$ and 70.4–80.4 km/s/Mpc for $v_{\rm CMB}$. Because all ten are computed from the same 39 galaxies, their differences are partly sampling noise: each estimator weights the same few distant galaxies differently, and so picks up their scatter differently. To separate the two effects, all ten estimators were recomputed on the same 4,000 bootstrap resamples of the sample (Table 6). This gives both the sampling noise of each estimator and the sampling noise of each *difference* between an estimator and the reference fit ($v_{\rm helio}$, ordinary least squares through the origin).

No estimator differs from the reference by more than 1.8 times the sampling noise of that difference. With 39 galaxies, therefore, the dependence of H_0 on the choice of estimator cannot be distinguished from sampling noise: the spread across estimators is best read as a visible symptom of the statistical uncertainty, rather than as an additional systematic error to be added on top of it (doing so would count much of the same noise twice). The heliocentric estimators agree closely (standard deviation 1.2 km/s/Mpc across the five), while the CMB-frame ones are more spread out (4.3 km/s/Mpc), consistent with the frame comparison in the Discussion below: in the CMB frame the nearby galaxies scatter much more, and the estimators that give them more weight move further.

3.3.3 Validation against the published MCP result

As an external check of the method itself, the weighted origin-forced fit of Section 3.3 was applied to the six MCP galaxies using the distances, distance uncertainties (symmetrized) and CMB-frame velocities published in Table 1 of Pesce et al. [2020], with the same 250 km/s peculiar-velocity uncertainty adopted there. It gives $H_0 = 75.4 \pm 3.1$ km/s/Mpc, compared

Frame	Estimator	H_0	SE	$\mathrm{SD}_{\mathrm{boot}}$	Δ	$\Delta/\mathrm{SD}(\Delta)$
v_{helio}	OLS, origin (reference)	74.11	1.59	1.64	—	—
	OLS, free intercept	73.39	2.05	2.26	−0.73	−0.5
	weighted, origin	74.33	2.67	2.40	+0.22	+0.2
	weighted, free intercept	71.92	3.52	2.49	−2.19	−1.4
	ODR	75.05	2.72	2.53	+0.93	+0.6
v_{CMB}	OLS, origin	75.70	2.20	2.12	+1.59	+1.4
	OLS, free intercept	71.99	2.68	2.99	−2.13	−0.9
	weighted, origin	78.89	3.82	3.65	+4.77	+1.7
	weighted, free intercept	70.44	4.51	3.31	−3.67	−1.5
	ODR	80.43	4.01	4.28	+6.32	+1.8

Table 6: The ten H_0 estimators of this analysis (km/s/Mpc): analytic standard error (SE), standard deviation over 4,000 paired bootstrap resamples ($\mathrm{SD}_{\mathrm{boot}}$), shift from the reference fit (Δ), and that shift in units of its own bootstrap standard deviation.

with the published 73.9 ± 3.0 km/s/Mpc: a difference of 0.4σ. Some difference is expected, since the published analysis uses each galaxy's full, asymmetric distance likelihood rather than a symmetrized error. Within its precision, the simple method used throughout this project reproduces a professional result.

3.4 Discussion

The four unweighted modern estimates (72.0–75.7 km/s/Mpc) lie close to the range spanned by current measurements, $H_0 \approx 67$–73 km/s/Mpc [Riess et al., 2022, Planck Collaboration, 2020], at its upper end, in stark contrast with Hubble's 1929 results (409–454 km/s/Mpc, depending on fit type and sample). Since the fitting procedure applied here is identical to the one used in Part 1, this large improvement cannot be attributed to a change in statistical method: it must instead reflect the far greater accuracy of modern distance measurements compared to those available in 1929. As in Part 1, the origin-forced fit lies close to, and in this case slightly above, the free-intercept fit for both velocity frames, consistent with the physically motivated form of Hubble's Law; the high R^2 values (0.94–0.97) show that a single linear relation accounts for the great majority of the variance in the velocities. The most notable result of this extension concerns the relationship between the two velocity frames. Repeating the same fits on the nearby galaxies alone ($r \leq 25$ Mpc, $n = 32$) shows that there, switching from heliocentric to CMB-frame velocities *worsens* the estimate: the origin-forced fit moves from 75.2 ± 5.2 to 85.7 ± 7.0 km/s/Mpc, and the free-intercept fit from 62.0 ± 12.5 to 55.2 ± 16.1 km/s/Mpc. These nearby-only fits are also far less precise ($R^2 = 0.18$–0.45, against 0.94–0.97 for the full sample), because at these distances peculiar velocities of a few hundred km/s are comparable to the expansion signal itself. Once the seven galaxies beyond 25 Mpc are added (the five MCP masers, at 51.5–132.1 Mpc, together with NGC 7619 and NGC 2986), this discrepancy nearly disappears: the heliocentric and CMB-frame fits now agree to within about 1.5 km/s/Mpc (73.4 vs. 72.0 for the free intercept; 74.1 vs. 75.7 forced through the origin). This is direct empirical confirmation of the hypothesis raised earlier: at short distances, local peculiar-velocity effects (dominated by the gravitational pull of the Virgo Cluster) are large enough to compete with, and even distort, the CMB-frame correction, but they become comparatively negligible once the sample is extended into the true Hubble flow, where cosmic expansion dominates galaxy kinematics.

How much do the six maser galaxies matter? Because the argument above rests on extending the sample to large distances, and because the five MCP galaxies (at 51.5–132.1 Mpc) are the points that reach farthest into the Hubble flow, it is worth asking directly how much the

maser-distance galaxies, specifically, are driving the result. Repeating the origin-forced fit with all six of them removed (NGC 4258 included; $n = 33$) gives $H_0 = 74.2 \pm 4.1$ km/s/Mpc for $v_{\rm helio}$ and $H_0 = 79.6 \pm 5.7$ km/s/Mpc for $v_{\rm CMB}$. Removing them therefore raises the central value by only 0.1 and 3.9 km/s/Mpc respectively, both within the standard error of the maser-free fit, but increases the uncertainty by a factor of about 2.6. The role of the masers is to anchor the fit with a long, precise lever arm, not to pull it toward a particular value. This revises the conclusion of earlier versions of this project (up to 1.1.0), in which removing the masers raised H_0 by about 10 km/s/Mpc. That shift came largely from CGCG 074-064: with its Fundamental Plane distance of 74.4 Mpc it implied $v/r \approx 93$ km/s/Mpc, making it an outlier among the non-maser galaxies, whereas it is now counted, with its geometric distance, among the maser galaxies. A complementary leave-one-out (jackknife) test, in which each galaxy is removed one at a time, shows that no single galaxy matters much either: removing any one of them shifts the origin-forced H_0 by at most 1.1 km/s/Mpc for $v_{\rm helio}$ (NGC 6264) and 1.4 km/s/Mpc for $v_{\rm CMB}$ (NGC 6323). The jackknife standard error (1.7 km/s/Mpc for $v_{\rm helio}$, 2.0 for $v_{\rm CMB}$) and the standard deviation of a bootstrap resampling of the full sample (20,000 resamples; 1.7 and 2.1 km/s/Mpc) agree with the analytic standard errors in Table 4 (1.6 and 2.2 km/s/Mpc), and the bootstrap 68% confidence intervals are $[72.5, 75.5]$ km/s/Mpc for $v_{\rm helio}$ and $[73.9, 77.7]$ km/s/Mpc for $v_{\rm CMB}$. Taken together, these checks indicate that the result is a property of the extended sample as a whole — specifically, of reaching into the true Hubble flow — rather than an artifact of one favorably-placed data point.

As a further check, the case of M31 (Andromeda) was examined using its modern, precisely calibrated distance. Despite the improved measurement, M31 still exhibits a negative heliocentric velocity ($v \approx -297$ km/s), consistent with the value already present in Hubble's original 1929 data. This confirms that the negative velocities observed among nearby galaxies are not a measurement artifact of 1929-era instrumentation, but a genuine dynamical effect: at short distances, gravitationally induced peculiar motion within the Local Group dominates over the recession caused by cosmic expansion — the same effect responsible for the frame discrepancy observed in the nearby subsample above. Finally, Figure 4 compares the origin-forced estimates with the literature. Using statistical errors only (the bootstrap standard deviation of each estimator combined with the quoted literature error), the reference estimate, 74.1 ± 1.6 km/s/Mpc, lies 0.4σ from the SH0ES value [Riess et al., 2022], 1.5σ from the CCHP TRGB value [Freedman et al., 2025], and 3.9σ above the Planck value [Planck Collaboration, 2020]. Across all ten estimators of Table 6, the distance from SH0ES never exceeds 1.6σ, while the distance from Planck ranges from 0.9σ to 3.9σ; the four free-intercept variants lie between 0.9σ and 2.6σ from Planck. This places the project's result on the local side of the tension, but it should not be read as independent support for it. The measurement is built entirely on local distance indicators. Its most important ingredient, the maser distances, are the same ones from which Pesce et al. [2020] derive their own local value of 73.9 ± 3.0 km/s/Mpc; NGC 4258 is the geometric anchor of the SH0ES Cepheid calibration; and the result moved by about 2.5 km/s/Mpc between versions 1.1.0 and 1.2.0 of this analysis simply by adopting the updated maser distances, a data-choice effect comparable to the statistical error itself. The comparison therefore reproduces, on a small scale, the same discrepancy between local and early-Universe determinations that defines the tension, rather than shedding light on its cause. It is also a useful reminder that "landing near a published value" and "being a precise, well-controlled measurement" are not the same claim, and this project is more confidently the former than the latter.

3.5 Limitations

Several limitations of this analysis should be acknowledged. First, the galaxy sample was not selected through a systematic or blind procedure, but by prioritizing objects with well-documented, high-quality distance measurements on NED; this could introduce a mild selection bias favoring galaxies that are easier to measure precisely. Second, the weighted fit in Section 3.3 is a

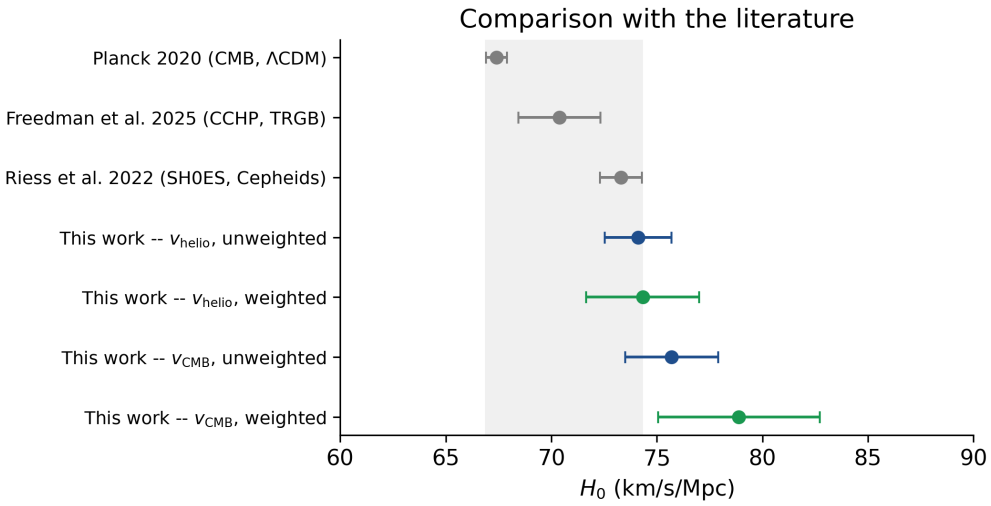


Figure 4: Comparison of this project's origin-forced H_0 estimates (unweighted and weighted, both velocity frames) with Planck 2020 [Planck Collaboration, 2020], the CCHP TRGB value of Freedman et al. 2025 [Freedman et al., 2025], and Riess et al. SH0ES 2022 [Riess et al., 2022]. Error bars on this project's estimates are the analytic standard errors. The shaded band spans from the lower end of the Planck uncertainty to the upper end of the SH0ES uncertainty.

partial answer to the unweighted-fit concern, but not a complete one: it uses representative, method-level fractional errors rather than the individual quoted uncertainty of each galaxy (for the maser galaxies in particular, the single 5% value is a simplification: the individual uncertainties published by Pesce et al. [2020] and Reid et al. [2019] range from about 1.5% for NGC 4258 to about 26% for NGC 6323), and its peculiar-velocity floor is a single isotropic number rather than a direction-dependent flow model; the persistently elevated χ^2/dof ($\gtrsim 1$ in every variant tried) is the fit's own admission of this simplification. The ODR check goes some way toward this by allowing for error on r as well as v, but it still relies on the same representative, method-level error budget rather than individual measurement uncertainties, so it sharpens the diagnosis without fully resolving it. Third, only two velocity frames (heliocentric and CMB) were tested; a more complete treatment of peculiar velocities would model the Virgo Cluster infall and other local flows explicitly, rather than relying on sample extension and a scatter floor to suppress their effect. Fourth, as the sensitivity analysis in Section 3.4 makes explicit, the precision of the result depends on a small number of long-baseline galaxies (above all the five MCP masers), and its central value depends on which published determination of their distances is adopted: updating from the earlier individual MCP distances to those of Pesce et al. [2020] moved the estimates by about 2.5 km/s/Mpc. This is not a flaw in the method, but it does mean the modern estimate here is less independent of any single distance-measurement technique than a large professional sample would be. Fifth, the Cepheid distances, taken from NED, do not come from a single homogeneous calibration: the four that were checked against the literature in version 1.2.1 had to be replaced (Section 3.1), and the other 16 have not been re-sourced from a single reference such as the final HST Key Project analysis [Freedman et al., 2001]. Because these galaxies all lie within about 21 Mpc, where the fit gives them little weight, this mainly affects the nearby-subsample fits and the Cepheid residuals rather than the full-sample estimate. Finally, the sample size ($n = 39$), while sufficient to obtain a stable estimate of H_0, remains small compared to the hundreds or thousands of objects used in professional determinations of the Hubble constant, so the results presented here should be understood as a stable proof-of-concept reproduction of a well-established method rather than a new independent constraint on H_0.

4 Conclusion

Reproducing Hubble's original 1929 analysis yields an estimate of H_0 roughly six times higher than the modern accepted value, primarily because the distances of the time were systematically too small — the result of a chain of errors, from the Cepheid zero point (about a factor of 2) to the misidentification of HII regions as the brightest stars in more distant galaxies — (compounded, to a lesser extent, by the inclusion of nearby galaxies whose peculiar motion is not representative of cosmic expansion). Applying the same statistical method to an extended modern sample of 39 galaxies, spanning distances from about 1.5 Mpc to 132 Mpc and using a combination of Cepheid, SBF, supernova, and geometric maser distances, yields unweighted H_0 estimates of 72.0–75.7 km/s/Mpc, with statistical errors of 1.6–2.7 km/s/Mpc: at the upper end of the range spanned by current measurements, closest to the local distance-ladder value, as expected for a measurement built entirely on local distance indicators. Jackknife and bootstrap resampling confirm the analytic uncertainties, and show that the six maser-distance galaxies set the precision of the result, by providing a long, geometric lever arm into the Hubble flow, without pulling its central value. Inverse-variance weighting and orthogonal distance regression move the estimate within 70.4–80.4 km/s/Mpc depending on velocity frame and estimator; a paired bootstrap shows that these moves are comparable to the estimators' own sampling noise (at most 1.8 times the noise of each difference), so with 39 galaxies the choice of method cannot be separated from statistical noise, and the spread across methods is best understood as a visible measure of that noise rather than as an extra error to add on top. The weighted method itself reproduces the published MCP value to within 0.4σ. If anything, the most important lesson of the project is that a defensible error budget, not just a best-fit number, is what turns a curve fit into a measurement — and that this includes errors in the analysis itself: versions 1.2.0 and 1.2.1 correct a double counting of the distance uncertainty in the ODR check and five individual distances that were mislabelled or outdated, and adopt updated maser distances that moved the result by about 2.5 km/s/Mpc, an effect comparable to the statistical error. Taken together, the comparison illustrates that the key advance separating Hubble's original measurement from the modern value of H_0 was not a change in statistical methodology, but nearly a century of improvement in the precision of the extragalactic distance scale — together with the importance of sampling galaxies far enough into the Hubble flow that local peculiar motions no longer bias the result, and of being honest about how much any single weighting choice can still move the answer.

A NED Query Details

Distances and velocities for the modern sample (Table 3) were retrieved from the NASA/IPAC Extragalactic Database (NED) in June 2026. Each object's common designation as listed in Table 3 (NGC, UGC, IC, or CGCG catalogue number) is also its NED object name and can be used directly in NED's "By Name" search to retrieve the same "Basic Data" and "Distances" entries used here, making the dataset independently reproducible without requiring a separate identifier list. The exceptions are the distances of the six maser galaxies, taken directly from Reid et al. [2019] (NGC 4258) and Table 1 of Pesce et al. [2020] (the five MCP galaxies), and five distances corrected after checking them against the literature: NGC 300 [Gieren et al., 2005], NGC 2403, NGC 3351, and NGC 3627 [Freedman et al., 2001], and NGC 4594 [McQuinn et al., 2016]. The file `mcp_pesce2020.csv` in the repository reproduces the distances, distance uncertainties and CMB-frame velocities of that table, used for the validation in Section 3.3.3.

Data Availability

All distance and redshift data used in this analysis are publicly available from the NASA/IPAC Extragalactic Database (NED) at `https://ned.ipac.caltech.edu` (see Appendix A for query details). Hubble's original 1929 dataset is reproduced from Hubble [1929]. The Python code used for the fits, uncertainty estimates, and sensitivity analysis, together with both datasets and all figures, is openly available at `https://github.com/Viserion137/hubble-constant-project` and archived on Zenodo; the DOI `https://doi.org/10.5281/zenodo.22260280` covers all archived versions and always resolves to the latest one.

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